Experiment I: Car-Following Model Calibration

1. Experiment Setup

- Briefly describe the experimental platform, driving simulator, and two-vehicle car-following configuration.
- Summarize the experimental procedure and the role of the leading and following vehicles.

2. Data Collection and Processing

2.1 Human-Driving Data Collection

- Describe the human-driving data collection procedure and summarize the collected dataset, including the driver/run information and the amount of data collected.
- Report the amount of usable trajectory data retained for calibration.

2.2 Data Processing Method

- Describe the processing method applied to the collected trajectory data.
- Explain how the leader–follower trajectory data are prepared for IDM calibration.

3. IDM Car-Following Model Calibration

3.1 Calibration Method

- Describe the selected calibration method, including the calibration criterion, numerical method, and main calibration procedure.
- Include the necessary mathematical formulations for the selected method.

3.2 Calibration Results

- Present the calibrated IDM parameters.
- Compare the measured human-driving trajectories with the corresponding model-generated trajectories.
- Report the calibration error and quantitatively evaluate the calibration performance.

4. Validation of Calibrated IDM

- Analyze the validation data by comparing the measured validation trajectories with the corresponding model-generated trajectories.
- Quantitatively evaluate the validation performance and report the validation error.

5. Discussion

5.1 Comparison between different data of vehicles

- Compare the calibrated IDM parameters obtained by the five groups.
- Identify the main differences among the five parameter sets and discuss what these differences indicate about the corresponding car-following characteristics.

5.2 Analysis of relevant scenarios

- Compare the validation results across the five groups using the provided validation datasets.
- Analyze representative validation scenarios in which different vehicle responses or model performances are observed.
- Relate the differences in the validation results to the corresponding calibrated IDM parameters.

5.3 Comparison with real-world IDM

Representative HV–HV and HV–AV car-following trajectories from the Third Generation Simulation Data (TGSIM) I-90/I-94 Moving Trajectories dataset will be provided.

- Calibrate the IDM parameters for the provided HV–HV and HV–AV trajectories using the same calibration method.
- Compare the calibrated IDM parameters from the real-world trajectories with those obtained from the SC^2I scaled testbed.
- Compare the corresponding calibration results and discuss the differences between the scaled-testbed and real-world car-following behavior.